Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I've received your dataset. I'll start by examining its structure, including the column names, data types, and a preview of the first few rows. This will help in understanding the dataset before proceeding with specific analytical questions. Let me load and inspect the data now.

The dataset contains 167,826 records and 9 columns. Here's a breakdown:

- **Country/Region**: The country or region of data collection.
- **UNICEF_Region**: The UNICEF region associated with the country (some missing values).
- **Indicator**: Description of the epidemiological measure (e.g., incidence rate).
- **Year**: The year of the recorded data.
- **Sex**: Gender classification (Both, Male, Female).
- **Age**: Age group classification (e.g., 0-14, 15-19).
- **Value**: The measured value (some appear as strings, like "<0.01").
- **Lower & Upper**: Confidence interval bounds (also contain string values like "<0.01").

Would you like me to clean or preprocess the data before proceeding with analysis? Also, what specific question would you like answered? [>-]

What is the HIV prevalence by age group and gender in 2023?

I will extract and analyze the HIV prevalence data for the year 2023, broken down by age group and gender. I'll ensure the **"Indicator"** column correctly represents prevalence rates and handle any data inconsistencies (e.g., "<0.01" values). Let me process the data now.

It looks like there are no direct matches for "prevalence" in the **Indicator** column for 2023. I'll recheck the available indicators to see how prevalence might be represented. Let me list the unique indicators in the dataset. [>-]

The dataset does not explicitly contain an "HIV prevalence" indicator. However, prevalence can be derived from:

$$\text{Prevalence Rate} = \frac{\text{Estimated number of people living with HIV}}{\text{Total population}} \times 100$$

Would you like me to calculate prevalence using this approach? If so, do we have total population data, or should I check for another method? [>-]

Follow the best approach to solve this problem based on the dataset.

Since the dataset includes the **"Estimated number of people living with HIV"**, I will use this metric to analyze HIV burden by **age group and gender in 2023**. However, without total population data, I will report absolute numbers instead of prevalence rates. Let me extract and summarize this information now.

HIV Burden by Age Group and Gender in 2023

The dataset provides the **estimated number of people living with HIV** rather than a direct prevalence rate. Below are the key findings:

- **Age Group 0-14**:
  - Both: **15,610**
  - Female: **15,450**
  - Male: **16,340**
- **Age Group 15-19**:
  - Both: **23,590**
  - Female: **23,390**
  - Male: **24,390**
- **Age Group 10-19**:
  - Both: **18,040**
  - Female: **14,120**
  - Male: **18,900**
- **Youngest Age Groups**:
  - **0-4**: Highest numbers in young children (Both: **18,710**)
  - **5-9**: Similar trends (Both: **18,640**)

This suggests that HIV burden is significant in adolescents (15-19) and young children (0-4). If you need further breakdowns or visualizations, let me know!

Here's the Python code used for this analysis:

python Copiar Editar

```python
# Filter for 2023 and relevant indicator
df_hiv_living_2023 = df[(df["Year"] == 2023) &
                        (df["Indicator"] == "Estimated number of people living with HIV")
```

```
                         (df["Indicator"] == "Estimated number of people living with HIV")

# Select relevant columns
df_hiv_living_2023 = df_hiv_living_2023[["Age", "Sex", "Value"]]

# Convert 'Value' to numeric, handling "<0.01" and other non-numeric values
df_hiv_living_2023["Value"] = pd.to_numeric(df_hiv_living_2023["Value"].str.replace("<",

# Aggregate by age group and gender (sum total cases)
hiv_living_summary_2023 = df_hiv_living_2023.groupby(["Age", "Sex"])["Value"].sum().reset_

# Display results
print(hiv_living_summary_2023)
```

Would you like any further analysis or a visualization of these results? [>_]

Envie uma mensagem para o Data Analyst